Limits
chapter 01

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Outline

- Limits of functions
- Algebraic limit

Introduction

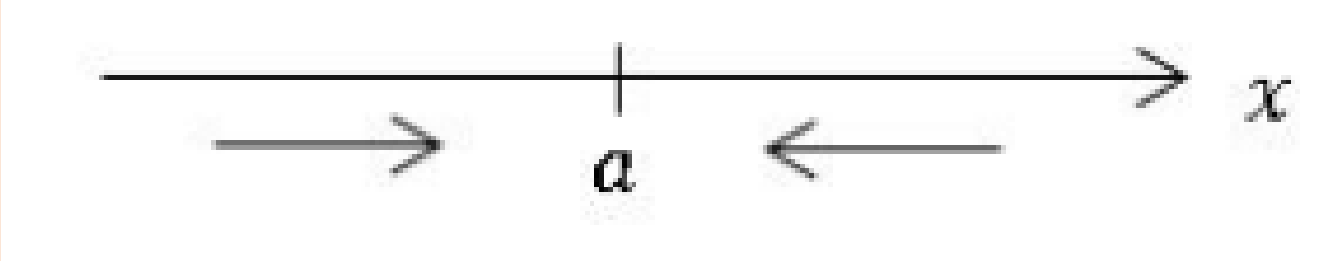
- We already know about the idea of a function; like $y = f(x)$
- Above function, y is a function of x and when x changes, y changes accordingly

$$Eg: - y = 3x + 1$$
$$F(x) = 3x + 1$$
$$F(1.9) = 6.7$$
$$F(2) = 7$$

- So that y can be determined for all given values of x
- Limit is the behavior of the function $f(x)$, when x approaches to a certain value "a"

The meaning of x goes to a

- We choose values of x that got closer and closer to $x = a$ and we plugged these into the function.
- Also consider the values of x that were on both the left and the right of $x = a$, but not equal to a.
- It is denoted by $x \to a$



- We can call x approaches to a

Example

Consider the function $|x-2|$. We get closer and closer to 2 on both lef and right sides but not equal to 2

$X < 2$	$X > 2$	$\lvert x-2 \rvert$
1.9	2.1	0.1
1.99	2.01	0.01
1.999	2.001	0.001
1.9999	2.0001	0.0001
1.99999	2.00001	0.00001

$|x-2|$ can be a very small number but never 0

Limits of functions

- Consider the function $f\ (x)$ when $x\ \rightarrow\ a$
- When x get closer and closer to a, we look at our table of function values and see what the function values are approaching.
- Using this we can guess the function value at $x\ =\ a$
- This process is called Taking limit of the function f at $x\ =\ a$. And it is denoted by $\lim\limits_{x \to a} f(x)$

Example

A father lends his son some money with the understanding that every month, the son pay back 10% of the balance remaining, then we have the following

Payment in first month $10\% \ = \frac{l}{10}$

Payment in second month= $l - \frac{l}{10} \ = \frac{9l}{10}$

Balance of the second month= $10\% \ of \ \left(\frac{9l}{10}\right)^2$

And soon after n monthly the amount paid each is $= \frac{l}{10}(1 + \frac{9}{10} + \left(\frac{9}{10}\right)^2 + \cdots .. + \left(\frac{9}{10}\right)^{n-1} = l\left(1 - \left(\frac{9}{10}\right)^{n-1}\right)$

This shows that the son never pays back the loan l in full.

But as he goes on paying, the amount paid back comes closer and closer to l

Example

- Consider the function $f\ (x)\ = \frac{2-2x^2}{1-x}$. We investigate the behavior of $f\ (x)$ near the point $x\ =\ 1$.
- When we substitute $x\ =\ 1$ in the given function we get,

$$f\ (0) = \frac{0}{0} \text{ which is undefined.}$$

- Now, we calculate the values of $f\ (x)$ when x get closer and closer to 1.

Example

$X < 1$	F(x)	$X > 1$	$F(x)$
0.9	3.8	1.1	4.2
0.99	3.98	1.01	4.02
0.999	3.998	1.001	4.002
0.9999	3.9998	1.0001	4.0002
0.99999	3.99998	1.00001	4.00002

According to the table, when x → 1 the function value gets closer to 4 from each side.

The limit of f (x) as x goes to 1 is 4.

Therefore, $\lim_{x \to 1} \frac{2-2x^2}{1-x} = 4$

Properties of Limits

The following properties of limits are important

1. If c is a constant , when f(x)=c ,$\lim_{x\to a} f(x) = c$
2. $\lim_{x\to a}\big(f(x) \pm g(x)\big) = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$
3. $\lim_{x\to a}\big(f(x) * g(x)\big) = \lim_{x\to a} f(x) * \lim_{x\to a} g(x)$
4. $\lim_{x\to a}\left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$; provided $\lim_{x\to a} g(x) \neq 0$
5. $\lim_{x\to a} \{f(x)\}^n = \{\lim_{x\to a} f(x)\}^{\,n} \;\; where\ n \in \mathbb{R}$
6. $\lim_{x\to a} Cf(x) = C\{\lim_{x\to a} f(x)\} \;\; where C \in \mathbb{R}$
7. $\lim_{x\to a} \{f(x)\}^{\frac{1}{n}} = \{\lim_{x\to a} f(x)\}^{\frac{1}{n}} \;\; where\ n \in \mathbb{Z}^+$

8. $\lim_{x \to a} x^n = a^n \; where \; n \in \mathbb{Z}^+$

Theorem

$$\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}; n \in \mathbb{R}$$

Questions

1. $\lim_{x\to -2}(2x^2+3x)$
2. $\lim_{x\to -1}\frac{x^2+2x-1}{x^3+5}$
3. $\lim_{x\to -5}(2x^2+1)(3x+1)$
4. $\lim_{x\to -1}\left(\frac{x^2+1}{x^2+2}\right)^5.$
5. $\lim_{x\to -2}\left(\frac{1}{\sqrt{x^2+5}}\right)$
6. $\lim_{x\to 2}\frac{x^2+3x-2}{x-1}$
7. $\lim_{x\to 0}\frac{x^2-6}{(x^2-1)(x+2)}$
8. $\lim_{x\to -3}\frac{x^3+27}{(x+3)}$
9. $\lim_{x\to 0}\frac{x^3+5x^2+x}{(x^3+7)}$
10. $\lim_{x\to 2}\frac{x^5-32}{x-2}$
11. $\lim_{x\to 3}\frac{x^3-3^3}{x-3}$
12. $\lim_{x\to 2}\frac{x^7-128}{x-2}$
13. $\lim_{x\to -2}\frac{x^5+32}{x+2}$
14. $\lim_{x\to 5}\frac{x^{1/6}-5^{1/6}}{x-5}$

Evaluation of algebraic limit

We use method of factorization here.

If f (x) and g(x) are polynomials and g(a)$\neq 0$. then we have,

$$\lim_{x \to a} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \frac{f(a)}{g(a)}$$

Now if f (a) = g(a) = 0, then we know that the limit is undefined.

But if f (a) = g(a) = 0, (x − a) is a factor of f (x) and g(x). So we can cancel this common factor and substitute x = a to the remaining expression.

When evaluating algebraic limits, first substitute the value which x getting at, to the function. If the result is undefined, use the method of factorization.

- We can use following results for simplifications.
- $x^2 - y^2 = (x - y)(x + y)$
- $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- $x^4 - y^4 = (x - y)(x + y)(x^2 + y^2)$

Example

- $\lim_{x \to 2} \frac{x^3 - 8}{x - 2}$
- $\lim_{x \to 2} \frac{x - 1}{2x^2 - 7x + 5}$
- $\lim_{x \to 2} \frac{x^2 - 9}{x - 3}$
- $\lim_{x \to 2} (x^2 - 9)\left(\frac{1}{x+3} + \frac{1}{x-3}\right)$

Limit of a function as x goes to infinity

- As x goes to infinity if a function approaches some value, then it is called the limit of the function as x goes to infinity.
- In this kind of problems, most of the time we take the x with highest power out from both numerator and denominator and cancel them out.
- then substitute $x = a$ to the remaining expression.

Note

Let c be a real number and k be a positive rational number. then,

$$\lim_{x \to \infty} \frac{c}{x^k} = 0$$

(a) $\lim_{x\to\infty} \frac{2x^3+x^2+2}{x^2+x+2}$

(b) $\lim_{x\to\infty} \frac{2x^3+x^2+2}{x^4+x^3+2}$

(c) $\lim_{x\to\infty} \frac{ax^2+bx+c}{cx^2+bx+a}$

(d) $\lim_{x\to\infty} \frac{\sqrt{1+x^3}-\sqrt{1+x}}{\sqrt{1+x^2}-\sqrt{1+x}}$

(e) $\lim_{x\to\infty} \frac{3x^2+4x+5}{x+1}$

(f) $\lim_{x\to\infty} \frac{3x+5}{x^2+1}$

(g) $\lim_{x\to\infty} \left(\sqrt{x^2+1}-x\right)$

Try

1. $\lim_{x\to 1}(5x^2+3x-4)$
2. $\lim_{x\to 3}\frac{x^2-x-6}{x-3}$
3. $\lim_{x\to 0}\frac{\sqrt{x+1}-\sqrt{1-x}}{x}$
4. $\lim_{x\to 2}\frac{x^2-4}{x^5-32}$
5. $\lim_{x\to \infty}\frac{x}{1-7x}$
6. $\lim_{x\to \infty}\frac{x^2-7}{(x+2)(x^2-1)}$
7. $\lim_{x\to 2}\frac{(x^2-4)}{(\mathrm{x}^2-1)(\mathrm{x}-2)}$
8. $\lim_{x\to -5}\frac{x^2-x-30}{x+5}$
9. $\lim_{x\to \infty}\frac{3x^2+4x+5}{4x^2+5x+6}$
10. $\lim_{x\to 3}\frac{x^5-243}{x^4-81}$
11. $\lim_{x\to 3}\frac{4-\sqrt{7+x^2}}{9-x^2}$
12. $\lim_{x\to \infty}\frac{5x^2+2x-1}{5x^2-3x+4}$

Thank you